Lab no. 7

Docker Setup and PostgreSQL Containers

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Task 1: Install Docker and Docker Compose

```
sania270804@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
sania270804@ubuntu:~$ docker compose version
Docker Compose version v5.6.1
sania270804@ubuntu:~$
```

Install Docker and Docker Compose.

Task 2: Run Two PostgreSQL Versions in Docker

```
sania270804@ubuntu:~$ docker pull postgres:9.6
9.6: Pulling from library/postgres
4b9a74ac6ea0: Pull complete
2d87c3a4038c: Pull complete
4e2c4902efbd: Pull complete
01c676df543a: Pull complete
1cb79db8a9e7: Pull complete
f955a6cf127b: Pull complete
1e3d335ef0b7: Pull complete
f62dc55c568d: Pull complete
e8622b8cb6f3: Pull complete
11087f2b0d87: Pull complete
d6d74bba3a57: Pull complete
f6bae7873dd7: Pull complete
8f7722dc50a7: Pull complete
874d4d2a09fd: Pull complete
Digest: sha256:caddd35b05cdd56c614ab1f674e63be778e0abdf54e71a7507ff3e28d4902698
Status: Downloaded newer image for postgres:9.6
docker.io/library/postgres:9.6
sania270804@ubuntu:~$ docker run  postgres:9.6
Error: Database is uninitialized and superuser password is not specified.
       You must specify POSTGRES_PASSWORD to a non-empty value for the
       superuser. For example, "-e POSTGRES_PASSWORD=password" on "docker run".

       You may also use "POSTGRES_HOST_AUTH_METHOD=trust" to allow all
       connections without a password. This is *not* recommended.

       See PostgreSQL documentation about "trust":
       https://www.postgresql.org/docs/current/auth-trust.html
sania270804@ubuntu:~$
```

Pull the PostgreSQL 9.6 image. Try to run the image without setting a password

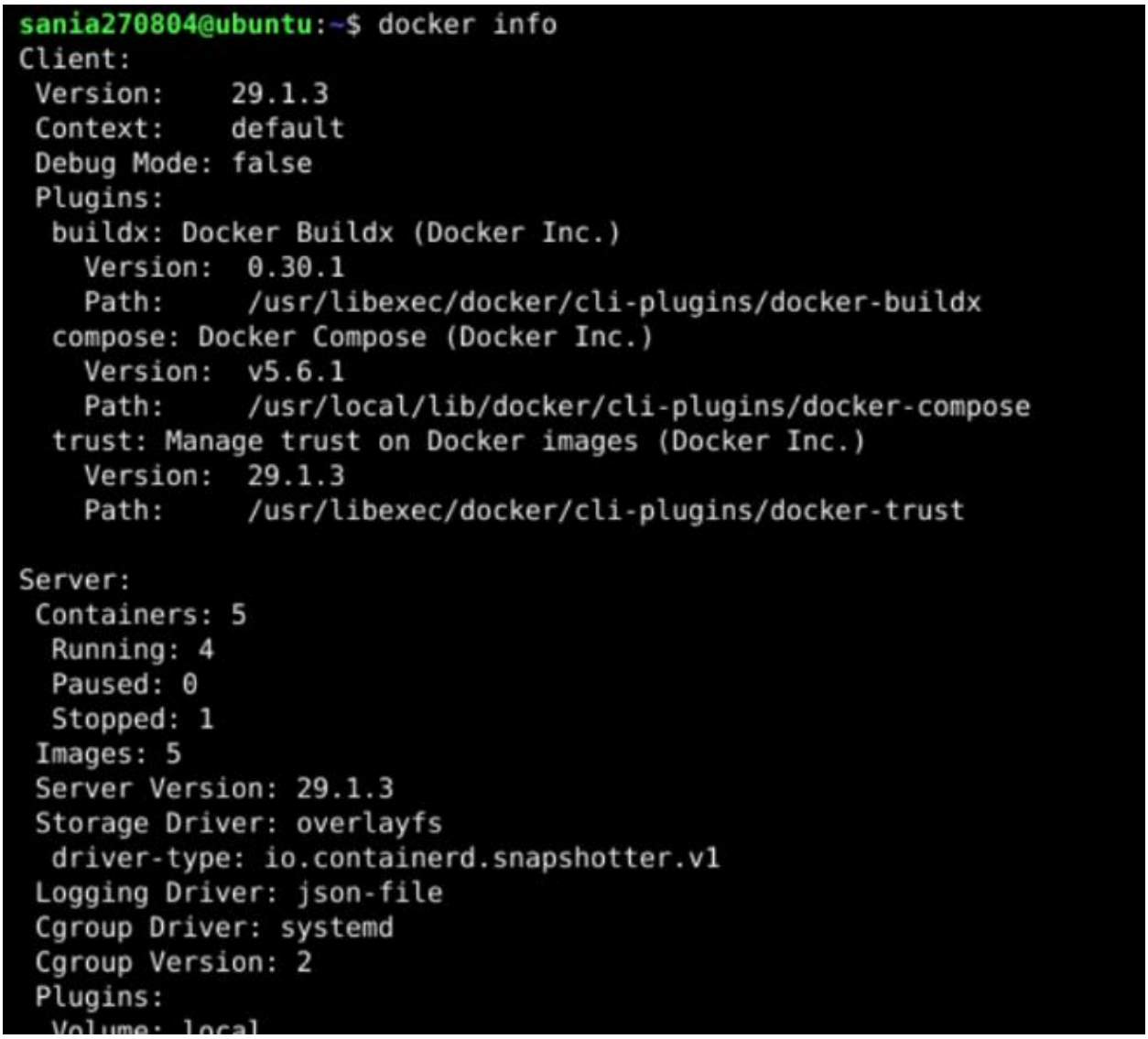
```
sania270804@ubuntu:~$ docker run -d -e POSTGRES_PASSWORD=password postgres:9.6
7e29138585bdcd381f9d8f59eee9976965ad95d878de0a502c12dd718cd027cb
sania270804@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED         STATUS
    PORTS                                                                                NA
MES
7e29138585bd   postgres:9.6             "docker-entrypoint.s…"   5 seconds ago   Up 4 seconds
    5432/tcp                                                                             st
range_leakey
651a396673d5   0b3563949d36             "/init"                  5 days ago      Up About a minut
e   3000/tcp, 0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp                               in
tellij-idea-docker-intellij-1
b3bcf179ef56   sonatype/nexus3:latest   "/opt/sonatype/nexus…"   5 days ago      Up 54 seconds
    0.0.0.0:8007->8007/tcp, [::]:8007->8007/tcp, 0.0.0.0:8006->8081/tcp, [::]:8006->8081/tcp   ne
xus
sania270804@ubuntu:~$
```

Start a new PostgreSQL 9.6 container with the password set. The -d flag runs it in the background and returns control to your terminal. List the running containers.

```
sania270804@ubuntu:~$ docker run -d -e POSTGRES_PASSWORD=password postgres:10.10
Unable to find image 'postgres:10.10' locally
10.10: Pulling from library/postgres
3106d02490d4: Pull complete
c5a7f905c8ec: Pull complete
16f2724dc303: Pull complete
ffc097878b09: Pull complete
756351b7a443: Pull complete
69802daaa561: Pull complete
88d1fc513b8f: Pull complete
f19250dffc5e: Pull complete
5c20c5b8227d: Pull complete
31babd3be108: Pull complete
0d24b08575ba: Pull complete
5a3f55930dd8: Pull complete
80369df48736: Pull complete
b18dd0a6efec: Pull complete
Digest: sha256:3263c8dad1b5aa6be80144786b18515910165ed771863d686c8b8fda30fea1ad
Status: Downloaded newer image for postgres:10.10
0cc379909b498df5a91f20111824ff37a6f826d5d7cfc57de8b665af7966019a
sania270804@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED              STATUS
         PORTS
   NAMES
0cc379909b49   postgres:10.10           "docker-entrypoint.s…"   4 seconds ago        Up 2 second
s        5432/tcp
   cool_thompson
7e29138585bd   postgres:9.6             "docker-entrypoint.s…"   About a minute ago   Up About a
minute   5432/tcp
   strange_leakey
```

Start a PostgreSQL 10.10 container in the background. List the running containers again.

Task 3: Inspect Docker and Manage Containers and Images



Display information about the Docker installation.

```
sania270804@ubuntu:~$ docker search ubuntu
NAME                      DESCRIPTION                                     STARS     OFFICIAL
ubuntu                    Ubuntu is a Debian-based Linux operating sys…   17878     [OK]
ubuntu/squid              Squid is a caching proxy for the Web. Long-t…   132
ubuntu/nginx              Nginx, a high-performance reverse proxy & we…   143
ubuntu/bind9              BIND 9 is a very flexible, full-featured DNS…   121
ubuntu/kafka              Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper          ZooKeeper maintains configuration informatio…   15
ubuntu/apache2            Apache, a secure & extensible open-source HT…   106
ubuntu/python             A chiselled Ubuntu rock with the Python runt…   34
ubuntu/prometheus         Prometheus is a systems and service monitori…   80
ubuntu/jre                Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet      Chiselled Ubuntu runtime image for ASP.NET a…   28
ubuntu/mysql              MySQL open source fast, stable, multi-thread…   74
ubuntu/dotnet-runtime     Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/postgres           PostgreSQL is an open source object-relation…   46
ubuntu/dotnet-deps        Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/redis              Redis, an open source key-value store. Long-…   26
ubuntu/jdk                Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq           Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/memcached          Memcached, in-memory keyvalue store for smal…   8
ubuntu/grafana            Grafana, a feature rich metrics dashboard & …   13
ubuntu/loki               Grafana Loki, a log aggregation system like …   3
ubuntu/traefik            Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/mimir              Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/php                PHP is a general-purpose scripting language …   10
ubuntu/grafana-agent      Ubuntu ROCK for Grafana Agent, an open-sourc…   0
sania270804@ubuntu:~$
```

Search Docker Hub for Ubuntu images.

```
sania270804@ubuntu:~$ docker images
                                                                  i Info →   U  In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest f760d82771b1       11.6GB         3.01GB    U
lab5:1.0                             7f6559bd3775       417MB          85.1MB
postgres:10.10                       3263c8dad1b5       310MB          77.7MB    U
postgres:9.6                         caddd35b05cd       294MB          73.6MB    U
sonatype/nexus3:latest               91a9af44af77       1.81GB         809MB     U
sania270804@ubuntu:~$ docker run -d ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
517b871c26ea: Pull complete
96aae9022e0e: Pull complete
e211d985152c: Download complete
Digest: sha256:065aa8626841f16057b78b027b464cd59c4fac36c204c72dcbf48d66f5fa8a96
Status: Downloaded newer image for ubuntu:latest
6b2c0b74475d03e47b9f1db1fbe693e1427ec8a0ff33953b44473cce9eadab24
sania270804@ubuntu:~$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS          PORTS
                                                                                NAMES
0cc379909b49   postgres:10.10         "docker-entrypoint.s…"   4 minutes ago   Up 4 minutes    5432/tcp
                                                                                cool_thompson
7e29138585bd   postgres:9.6           "docker-entrypoint.s…"   5 minutes ago   Up 5 minutes    5432/tcp
                                                                                strange_leakey
651a396673d5   0b3563949d36           "/init"                  5 days ago      Up 7 minutes    3000/tcp, 0.0.
0.0:3001->3001/tcp, [::]:3001->3001/tcp                                         intellij-idea-docker-intellij-1
b3bcf179ef56   sonatype/nexus3:latest "/opt/sonatype/nexus…"   5 days ago      Up 52 seconds   0.0.0.0:8007->
8007/tcp, [::]:8007->8007/tcp, 0.0.0.0:8006->8081/tcp, [::]:8006->8081/tcp   nexus
sania270804@ubuntu:~$ docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS                      P
ORTS                                                                                    NAMES
6b2c0b74475d   ubuntu                 "/bin/bash"              12 seconds ago   Exited (0) 10 seconds ago
                                                                                        festive_hamilton
0cc379909b49   postgres:10.10         "docker-entrypoint.s…"   4 minutes ago    Up 4 minutes                5
432/tcp                                                                                 cool_thompson
7e29138585bd   postgres:9.6           "docker-entrypoint.s…"   6 minutes ago    Up 6 minutes                5
432/tcp                                                                                 strange_leakey
```

List local images, start an Ubuntu container, and compare the two container lists. Give the new Ubuntu container a moment to exit before running docker ps.

```
sania270804@ubuntu:~$ docker restart strange_leakey
strange_leakey
sania270804@ubuntu:~$ docker stop strange_leakey
strange_leakey
sania270804@ubuntu:~$ docker rm strange_leakey
strange_leakey
sania270804@ubuntu:~$ docker rmi postgres:9.6
Error response from daemon: conflict: unable to delete postgres:9.6 (must be forced) - container 53e165c86874 i
s using its referenced image caddd35b05cd
sania270804@ubuntu:~$ docker rmi -f postgres:9.6^C
sania270804@ubuntu:~$ docker rmi postgres:9.6
Error response from daemon: conflict: unable to delete postgres:9.6 (must be forced) - container 53e165c86874 i
s using its referenced image caddd35b05cd
sania270804@ubuntu:~$ docker rmi -f postgres:9.6
Untagged: postgres:9.6
```

Restart, stop and remove the PostgreSQL 9.6 container, then remove its image tag.